

CS101

Introductions

1/13

Welcome to CS101!

Please work on

Getting to Know You

Agenda

1. Intros
2. Class Expectations + Teacher/Student Contract
3. Brief Lecture
4. Codespaces Setup + Read Lab1
5. Touch Typing Initial Test (time permitting)

10 min break at ~5pm

Class Instructor: Brent Magnusson

- Mr. Magnusson or Mr. M (he/him)
- Grew up near Seattle
- BA physics, economics | MS mechanical engineering
- Worked as:
 - Mechanical Engineer
 - Electrical Engineer
 - Software Test Engineer
 - Data Scientist
 - Software Engineer/Developer
- Washington State Teaching credential in HS Math, Physics, Computer Science

Intros

- Name + (optional) pronouns
- School and year in school
- Non-programming non-academic fun thing you like to do

Class Expectations and Teacher/Student Contract

Phone Policy

Phones collected at start of class (4pm)

Can use during break (or 2FA)

Why?

- Need sustained attention to problem solve
- Phones are *designed* to distract attention

What is Computer Science?

What is Computer Science?

Computer

- 1600's - 1940's "One who computes" mathematical operations
- 1940's - Machine computers invented

Science

- Systematic study of the world through observation and experimentation

Wiki Definition

- "Computer science is the study of computation, information, and automation."

Job Titles for Those Who Code

Computer Scientist: Studies different ways to compute or solve problems

Software Developer/Engineer: Writes code to solve real-world problems

Data Scientist: Analyzes/visualises data, trains AI systems using data

Can use coding skills for...

Automating repetitive tasks, analyzing data

- Finance
- Healthcare
- Any research job
- Data journalism
- Public policy
- Marketing
- Environmental science

Ways to learn code

- Online tutorials
- Videos
- Project based (on-the-job)
- Asking peers/coworkers
- Studying for Exams
- Reading code of libraries/plugins
- Using AI as a tutor/teacher

CS101 Method of Instruction

- Minimal Lecture
- Spend (most) of class time:
 - Individual learning (bring headphones)
 - Working on Labs (individual or small groups)
 - Working on Homeworks/Projects (individual)
 - Periodic Exam prep/taking
 - Touch Typing Practice

Computer Setup

Course Resources

Bookmark these:

- canvas.arcadia.edu
- github.com/icanhazcodeplz/CS101

Codespaces Setup

1. Make account at github.com
 - a. I recommend using a personal email
 - b. Username can be professional sounding, or not!
2. Go to: <https://cs50.dev>
 - a. Click “Log in”
 - This (should) create a codespaces environment for you (will take awhile)
3. UI Configuration (optional)
 - a. File -> Auto Save
 - b. View -> Appearance -> Menu Bar
 - c. Right Click “EXPLORER” -> Open Editors

Your Task

- In codespaces, create new folder called “lab1” with a file named “shell_basics.sh” inside it
- Read through “Lab1 - Shell Script”
 - Canvas → Assignments → Lab 1 - Shell Script (links to GitHub)

Touch Typing

- Write at the speed of thought!
- Boost confidence and productivity
- I practice regularly
 - Typing is fun now

Touch Typing Initial Test

Extra Credit up to 4% of final grade based on improvement over semester

1. Make account at [typing.com](https://www.typing.com)
2. Canvas → Assignments → Touch Typing Setup and Initial Test

“Homework”

- Spend *at least* 30 min learning about Topic 1 or doing Lab1

Next Class 1/15

- Syllabus
- Continue Topic 1 or start Topic 2